



LOGICLABS TECHNOLOGIES

Amazon Web Services

Elastic Block Store

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Elastic Block Store - Linux

- One EC2 Machine can have multiple Volumes also & Additional volume is known as Elastic Block Store (EBS)
- Creating Linux EC2 Machine using the multiple Volumes
- Go to Add Storage
- Click on Add New Volume
- Check Delete on Termination
- Launch the Instance

Elastic Block Store - Linux

- Check Volumes

Elastic Block Store



Volumes

- Mount the volume
- Connect Linux EC2 machine using putty

- **Check disk space usage**

`df -hT`

- **Check available disk device**

`lsblk`

Note down the name of the added ebs volume (xvdb)

Elastic Block Store - Linux

- **Information about device**

```
sudo file -s /dev/xvdb
```

- **Create file system on the volume**

```
sudo mkfs -t ext4 /dev/xvdb
```

- **Create Directory**

```
mkdir test
```

- **Mount the Volume**

```
sudo mount /dev/xvdb test
```

Elastic Block Store - Linux

- **Change directory**

```
cd test
```

- **Create files**

```
sudo touch file1 file2 file3
```

- **Check the files**

```
ls
```

- **Check disk space usage**

```
df -hT
```

Elastic Block Store - Linux

- **Change directory**

`cd`

- **Unmount the EBS volume**

`sudo umount test`

- **Detach Volume**

Actions



Detach Volume

Click on Detach

Elastic Block Store - Linux

- Create another Linux ec2 machine with root volume in same availability zone in which we have our EBS volume
- Attach Volume
- Select the Volume which we want to attach with EC2 Machine

Actions



Attach Volume

- Select the instance
- Click on attach volume

Elastic Block Store - Linux

- Now connect the second ec2 machine using putty

- **Check disk space usage**

`df -hT`

- **Check available disk device**

`lsblk`

- Note down the name of the added ebs volume (xvdf)

- **Information about device**

`sudo file -s /dev/xvdf`

Elastic Block Store - Linux

- **Create directory**

```
mkdir training
```

- **Mount the Volume**

```
sudo mount /dev/xvdf training
```

- **Change directory**

```
cd training
```

- **Check the files**

```
ls
```

Elastic Block Store - Windows

- Create Windows EC2 Machine with EBS Volume
- Check the volumes

Elastic Block Store



Volumes

- Connect the instance using remote desktop connection
- Click on start button
- Open server manager

Elastic Block Store - Windows

- Click on files and storage services
- Click on disks
- Select the disk & right click
- Click on bring online
- Click on yes
- Right click on the disk
- Click on initialize
- Click on yes

Elastic Block Store - Windows

- Right click on the disk
- Click on new volume
- Click on next (Multiple Times)
- Click on create
- Now check the windows system
- Open D Drive & create one file

Elastic Block Store - Windows

- Unmount the volume from ec2 machine
- Go to Server manager
- Click on files and storage services
- Click on disks
- Right click on EBS disk
- Click on Take Offline
- Click on Yes

Elastic Block Store - Windows

- Detach the volume

- Select the volume

Actions



Detach Volume

- Click on Detach

- Create windows ec2 machine in the same availability zone in which we have our EBS volume

- Attach the EBS volume with instance

Actions



Attach Volume

Elastic Block Store - Windows

- Select the instance
- Click on attach volume
- Connect the new windows ec2 machine using Remote desktop connection
- Open server manager
- Click on file and storage services
- Click on disks

Elastic Block Store - Windows

- Our ebs disk is showing as offline. Make that disk online
- Select the ebs disk
- Right click on ebs disk & click on being online
- Click on Yes
- Now check this PC
- Open D Drive
- [Click here](#) For More Details About Volume Type

Replace Root Volume

- **Replace Root Volume in AWS - Launch State:** [Click Here](#)
- **Replace Root Volume in AWS - Snapshot:** [Click Here](#)
- **Replace Root Volume in AWS - AMI:** [Click Here](#)
- **Device names on Linux instances:** [Click Here](#)
- **Device names on Windows instances:** [Click Here](#)



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